Mobile computing

Chapter 1: Applications

Outlines

- Short History of wireless communication
- Wireless Network Benefits
- Frequencies for Radio Transmission
- Signal Propagation
- Multiplexing and Modulation

A short history of wireless communication

- Heinrich Hertz (1857–94) was the first to demonstrate the wave character of electrical transmission through space (1886).
- Today the unit Hz reminds us of this discovery.
- Nikola Tesla (1856–1943) soon increased the distance of electromagnetic transmission.
- The first network in Germany was the analog A-Netz from 1958, using a carrier frequency of 160 MHz transmission.
- Nineteen ninety-eight marked the beginning of mobile communication using satellites with the Iridium system .

Benefits of wireless network

- Mobility: A wireless communication system allows users to access information beyond their desk and conduct business from anywhere without having a wire connectivity.
- Reachability: Wireless communication systems enable people to be stay connected and be reachable, regardless of the location they are operating from.
- Simplicity: Wireless communication systems are easy and fast to deploy in comparison to cabled networks.
- Maintainability: In a wireless system, you do not have to spend too much cost and time to maintain the network setup.

Frequencies for radio transmission

- Radio transmission can take place using many different frequency bands.
- Each frequency band exhibits certain advantages and disadvantages.
- Radio transmission starts at several kHz, the very low frequency (VLF) range.
- Waves in the low frequency (LF) range are used by submarines, because they can penetrate water and can follow the earth's surface.
- Some radio stations still use these frequencies, e.g., between 148.5 kHz and 283.5 kHz in Germany.

Cont..

- The medium frequency (MF) and high frequency (HF) ranges are typical for transmission of hundreds of radio stations either
  - as amplitude modulation (AM) between 520 kHz and 1605.5 kHz,
  - as short wave (SW) between 5.9 MHz and 26.1 MHz, or
  - as frequency modulation (FM) between 87.5 MHz and 108 MHz.
- Super high frequencies (SHF) are typically used for
  - directed microwave links (approx. 2–40 GHz) and
  - fixed satellite services in the
    - C-band (4 and 6 GHz),
    - Ku-band (11 and 14 GHz),
    - or Ka-band (19 and 29 GHz).

Cont..

- The next step into higher frequencies involves optical transmission,
  - which is not only used for fiber optical links but also for wireless communications.

- Infra red (IR) transmission is used for directed links, e.g., to connect different buildings via laser links.

Signal propagation

- Signals are the physical representation of data.
- Users of a communication system can only exchange data through the transmission of signals.

- Like wired networks, wireless communication networks also have senders and receivers of signals.
- However, in connection with signal propagation, these two networks exhibit considerable differences.

Cont..

- In wireless networks, the signal has no wire to determine the direction of propagation, whereas signals in wire transmission is depend on
- wire type,
- length and
- quality

between sender and receiver.

- For wireless transmission, this predictable behavior is only valid in a vacuum, i.e., without matter between the sender and the receiver.

Cont..

- The situation in wireless signal transmission would be as follows;
- Transmission range: Within a certain radius of the sender transmission is possible.
- Detection range: Within a second radius, detection of the transmission is possible, i.e., the transmitted power is large enough to differ from background noise. However, the error rate is too high to establish communication.
- Interference range: Within a third even larger radius, the sender may interfere with other transmission by adding to the background noise.

Cont..

- Depending on the frequency, radio waves can also penetrate objects.
- Generally the lower the frequency, the better the penetration.
- Long waves can be transmitted through the oceans to a submarine while high frequencies can be blocked by a tree.

- A receiver will not be able to detect the signals, but the signals may disturb other signals.
- Radio waves can exhibit three fundamental propagation behaviors depending on their frequency.

Cont..

I. Ground wave (<2MHz):

- with low frequencies follow the earth's surface and can propagate long distances.
- These waves are used for, e.g., submarine communication or AM radio.

II. Sky wave (2–30 MHz):

- Many international broadcasts and amateur radio use these short waves that are reflected at the ionosphere.
- This way the waves can bounce back and forth between the ionosphere and the earth's surface, travelling around the world.

Cont..

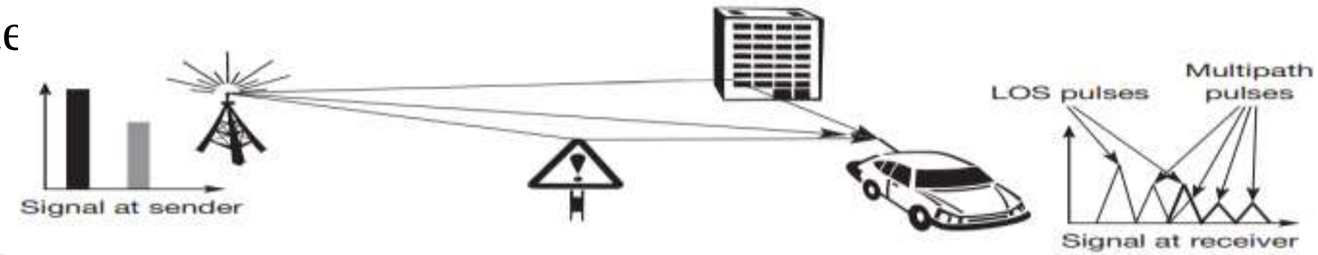
III. Line-of-sight (>30 MHz):

- Mobile phone systems, satellite systems, cordless telephones etc.
- use even higher frequencies.

- The emitted waves follow a (more or less) straight line of sight.
- This enables direct communication with satellites (no reflection at the ionosphere) or microwave links on the ground.

Multi-path propagation

- Radio waves emitted by the sender can either travel along a straight line, or they may be reflected at a large building, or scattered at smaller obstacles also called Multi-path propagation.
- The effect caused by multi-path propagation is called **delay spread**.
- Typical values for delay spread are approximately 3 µs in cities, up to 12 µs can be observed.
- Global System for Mobile Communication(GSM), for example, can tolerate up to 16 µs of delay spread, i.e., almost a 5 km path difference



Multiplexing

- Multiplexing describes how several users can share a medium with minimum or no interference.
- For wireless communication, multiplexing can be carried out in four dimensions:
  - space,
  - time,
  - frequency,
  - and code.

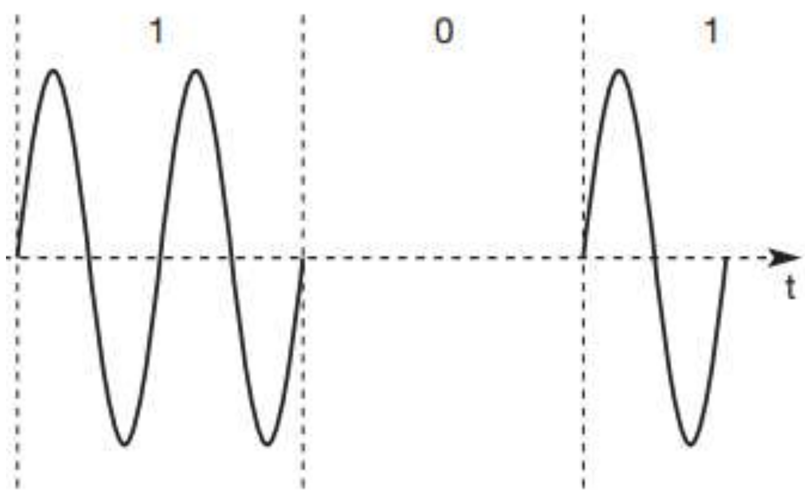
Modulation

- amplitude At ,
- frequency ft , and
- phase φt which may be varied in accordance with data or another modulating signal are the three basic components in digital modulating signals.
- In wireless networks, however, digital transmission cannot be used. Here, the binary bit-stream has to be translated into an analog signal first.
- The three basic methods for this translation are
  i. Amplitude shift keying (ASK),
  ii. Frequency shift keying (FSK), and
  iii. Phase shift keying (PSK).

I. Amplitude shift keying

I. Amplitude shift keying (ASK);

- the most simple digital modulation scheme.
- The two binary values, 1 and 0, are represented by two different amplitudes.
- In the example, one of the amplitudes is 0 (representing the binary 0).
- This simple scheme only requires low bandwidth.



Amplitude shift keying (ASK)

II. Frequency shift keying

- A modulation scheme often used for wireless transmission is frequency shift keying (FSK) .
- The simplest form of FSK, also called binary FSK (BFSK), assigns one frequency f1 to the binary 1 and another frequency f2 to the binary 0.
- To avoid sudden changes in phase, special frequency modulators with continuous phase modulation, (CPM) can be used.

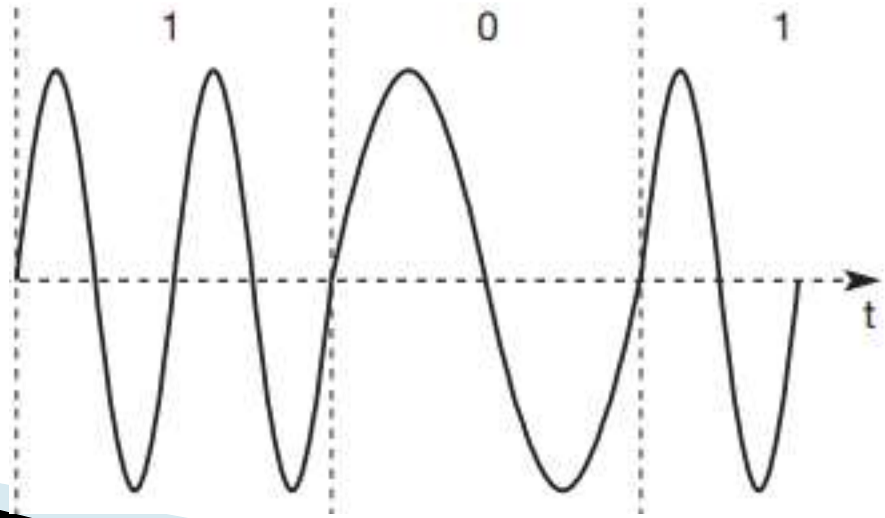
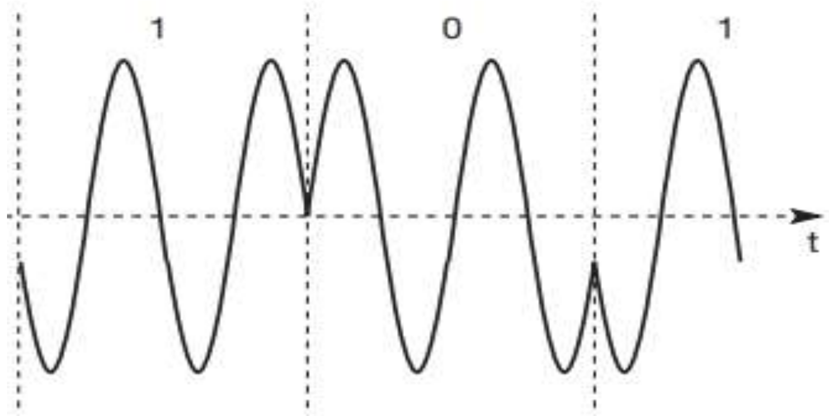


Figure 2.24 Frequency shift keying (FSK)

III. Phase shift keying

- III. Phase shift keying (PSK)
- uses shifts in the phase of a signal to represent data.
- In the Figure bellow shows a phase shift of 180° or π as the 0 follows the 1 (the same happens as the 1 follows the 0).
- This simple scheme, shifting the phase by 180° each time the value of data changes, is also called binary PSK (BPSK).



Phase shift keying (PSK)

THANK YOU!!!

Mobile computing

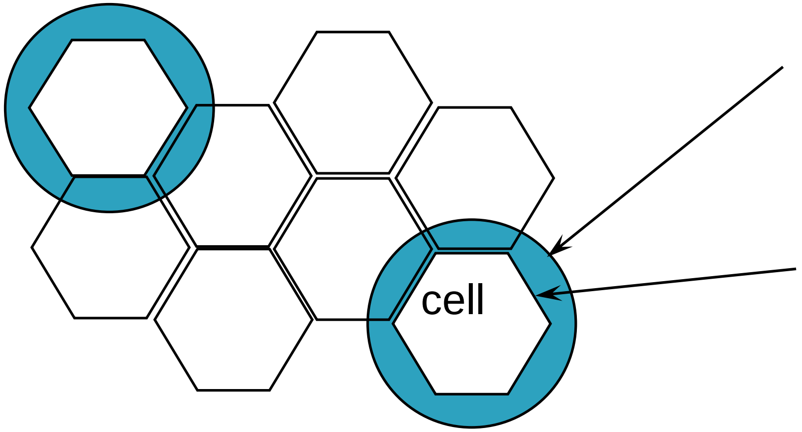
Chapter 2: Medium Access Control

Outlines

- Introduction
- Motivation for a Specialized MAC
- Space Division Multiple Access (SDMA)
- Frequency Division Multiple Access (FDMA)
- Time Division Multiple Access (TDMA)
- Code Division Multiple Access (CDMA)

Introduction

- The frequency spectrum allocated to wireless communications is very limited, so the cellular concept was introduced to reuse the frequency.
- In a cellular network, a service coverage area is divided into smaller hexagonal areas referred to as cells.
- Each cell contains a base station which transmits signals to and receives signals from the mobile stations in its cell.



Cont..

- The coverage area of a cell depends on many factors:-
  - The transmitting power of the base station
  - The transmitting power of the mobile station
  - Obstructing buildings in the cell
  - The height of base station antennas

- **Air interface Access Techniques**

- Typically, many simultaneous calls takes place in a given cell.
- These calls need to share the portion of the radio spectrum that is allocated to the cellular service provider.

Motivation for MAC

- The sender starts sending but a collision happens at the receiver due to a second sender.
- The sender detects no collision and assumes that the data has been transmitted without errors.
- but a collision might actually have destroyed the data at the receiver.
- Collision detection is very difficult in wireless scenarios as the transmission power in the area of the transmitting antenna is several magnitudes higher than the receiving power.

Cont..

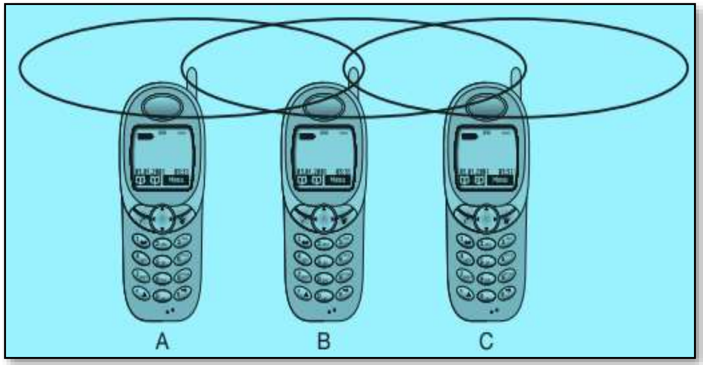
- So, this very common MAC scheme from wired network fails in a wireless scenario.
- The following sections show some more scenarios where schemes known from fixed networks fail.
- The scenarios where schemes known from fixed networks fail are.

  I. Hidden and exposed terminals

  II. Near and far terminals

i. Hidden and exposed terminals

- Consider the scenario with three mobile phones as shown in the Figure.



- The transmission range of A reaches B, but not C (the detection range does not reach C either).
- The transmission range of C reaches B, but not A.
- Finally, the transmission range of B reaches A and C, i.e., A cannot detect C and vice versa.
- A starts sending to B, C does not receive this transmission.
- C also wants to send something to B and senses the medium.
- The medium appears to be free, the carrier sense fails.

Cont..

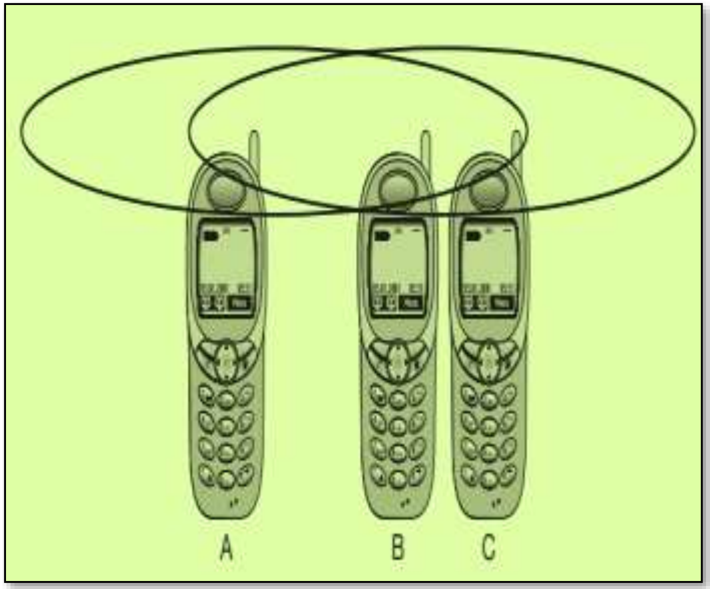
- C also starts sending causing a collision at B.
- But A cannot detect this collision at B and continues with its transmission.
- A is hidden for C and vice versa.
- While hidden terminals may cause collisions, the next effect only causes unnecessary delay.
- Now consider the situation that B sends something to A and C wants to transmit data to some other mobile phone outside the interference ranges of A and B.
- C senses the carrier and detects that the carrier is busy (B's signal).
- C postpones its transmission until it detects the medium as being idle again.

ii. Near and far terminals

- Consider the situation as shown in Figure A and B are both sending with the same transmission power.

- As the signal strength decreases proportionally to the square of the distance, B's signal drowns out A's signal.

Near and Far Terminals



- As a result, C cannot receive A's transmission.

Cont..

- Now think of C as being an arbiter for sending rights.
- In this case, terminal B would already drown out terminal A on the physical layer.
- C in return would have no chance of applying a fair scheme as it would only hear B.
- The near/far effect is a severe problem of wireless networks using CDM.

Cont..

- All signals should arrive at the receiver with more or less the same strength.
- Otherwise a person standing closer to somebody could always speak louder than a person further away.

- Even if the senders were separated by code, the closest one would simply drown out the others.
- Precise power control is needed to receive all senders with the same strength at a receiver.

Space Division Multiple Access (SDMA)

- is used for allocating a separated space to users in wireless networks.
- A typical application involves assigning an optimal base station to a mobile phone user.
- The mobile phone may receive several base stations with different quality.
- Typically, SDMA is never used in isolation but always in combination with one or more other schemes.

- The basis for the SDMA algorithm is formed by cells and sectored antennas.
- Single users are separated in space by individual beams. This can improve the overall capacity of a cell

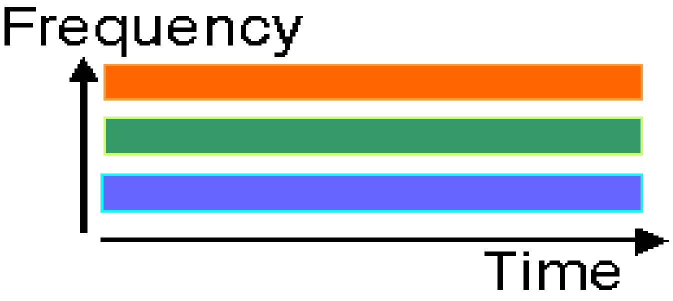
Frequency division multiple access (FDMA)

- comprises all algorithms allocating frequencies to transmission channels according to the frequency division multiplexing (FDM).
- Allocation can either be fixed or dynamic.
- Channels can be assigned to the same frequency at all times, i.e., pure FDMA, or change frequencies according to a certain pattern,
- i.e., FDMA combined with TDMA.

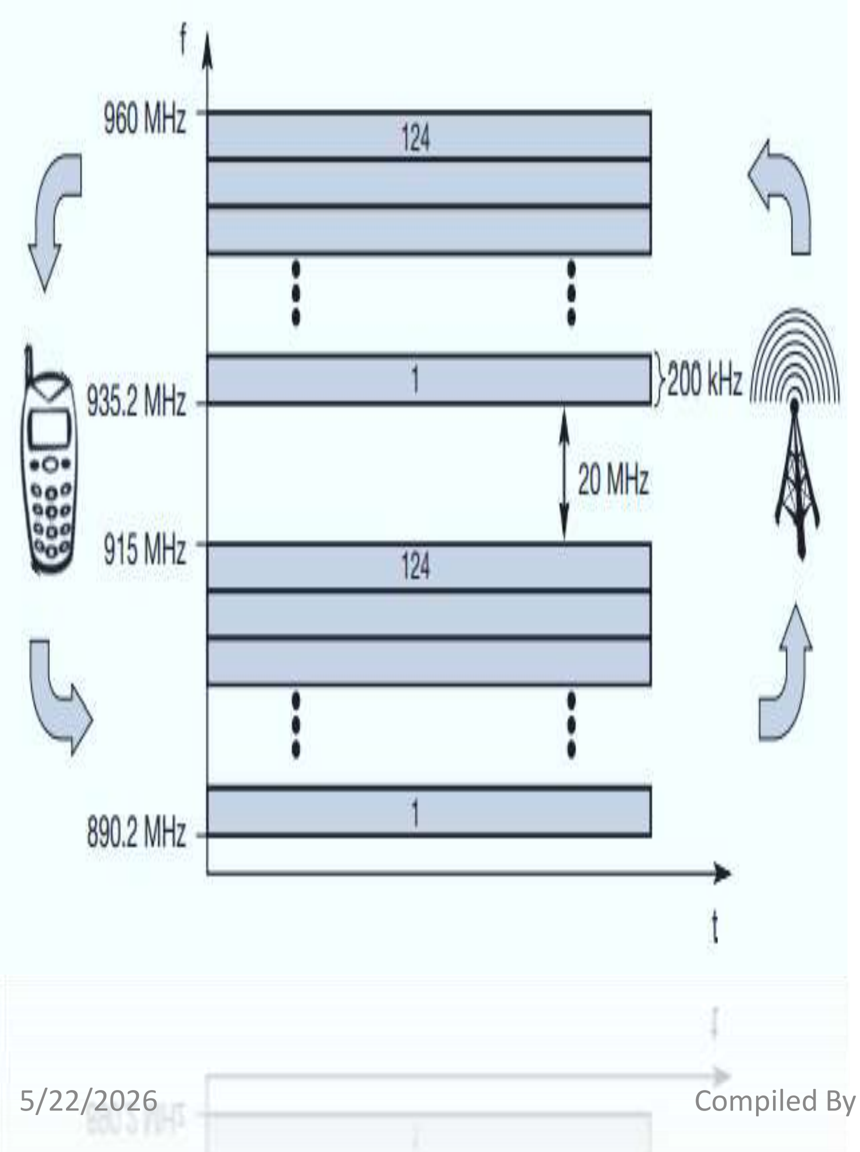
- The latter example is the common practice for many wireless systems to avoid narrowband interference at certain frequencies, known as frequency hopping.

Cont..

- Sender and receiver have to agree on a hopping pattern, otherwise the receiver could not adjust to the right frequency.
- Hopping patterns are typically fixed, at least for a longer period.
- The fact that it is not possible to arbitrarily jump in the frequency space is one of the main differences between FDM schemes and TDM schemes.
- Furthermore, FDM is often used for simultaneous access to the medium by base station and mobile station in cellular networks.
- Every user has his/her own frequency channel



Cont..



- The two frequencies are also known as
- **uplink**, i.e., from mobile station to base station or from ground control to satellite, and
- **downlink**, i.e., from base station to mobile station or from satellite to ground control.

Time Division Multiple Access (TDMA)

- Compared to FDMA, time division multiple access (TDMA) offers a much more flexible scheme.
- listening to different frequencies at the same time is quite difficult.

but

- listening to many channels separated in time at the same frequency is simple.
- Almost all MAC schemes for wired networks work according to this principle.

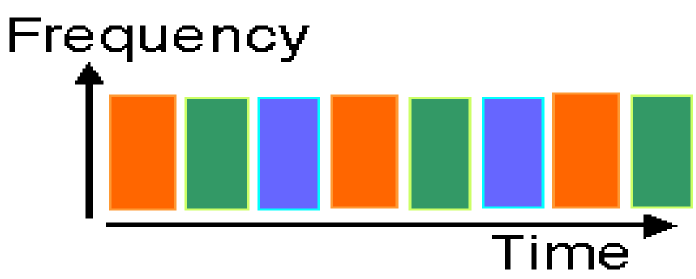
Cont..

- Now synchronization between sender and receiver has to be achieved in the time domain.
- Again this can be done by using a fixed pattern similar to FDMA techniques.

- Dynamic allocation schemes require an identification for each transmission as this is the case for typical wired MAC schemes.
- MAC addresses are quite often used as identification.

Cont..

- This enables a receiver in a broadcast medium to recognize if it really is the intended receiver of a message.
- Fixed schemes do not need an identification,
- but
  - are not as flexible considering varying bandwidth requirements.
- Users share the same bandwidth but transmit one after the other.



Code Division Multiplexing (CDM)

- use exactly these codes to separate different users in code space and to enable access to a shared medium without interference.
- The main problem is how to find
  - "good" codes and
  - how to separate the signal from noise generated by other signals and the environment.
- The code directly controls the chipping sequence.
- But what is a good code for CDMA?

Cont..

- A code for a certain user should have a good autocorrelation and should be orthogonal to other codes.
- Orthogonal in code space has the same meaning as in standard space (i.e., the three dimensional space).

- CDMA can be used in combination with FDMA/TDMA access schemes to increase the capacity of a cell.
- In contrast to other schemes, CDMA has the advantage of a soft handover and soft capacity.

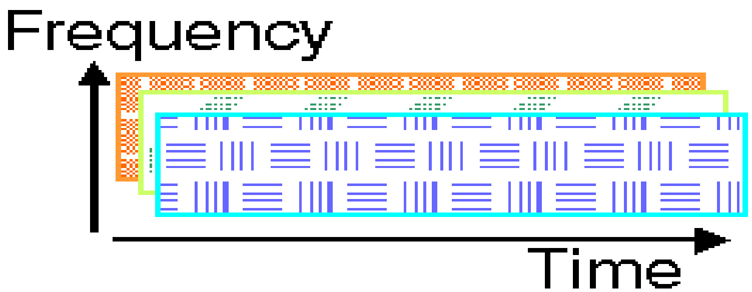
Cont..

- Handover describes the switching from one cell to another, i.e., changing the base station that a mobile station is connected to.
- Soft handover means that a mobile station can smoothly switch cells.

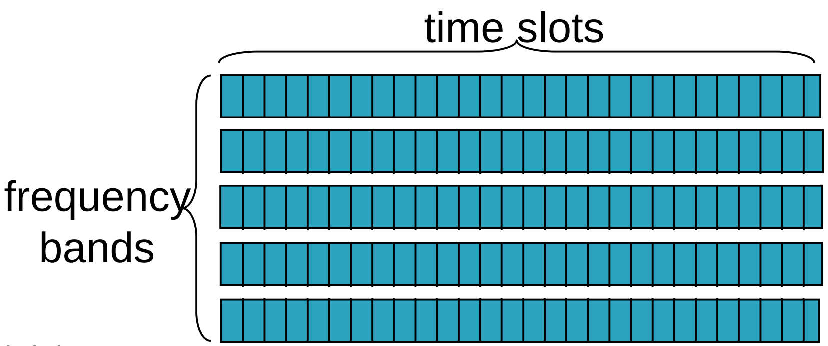
- This is achieved by communicating with two base stations at the same time.
- CDMA does this using the same code and the receiver even benefits from both signals.

Cont..

- User signals overlap in frequency and time.
- Orthogonally of waveforms is used to separate user signals.
- Most cellular systems today use one of two broad approaches for sharing radio spectrum.



- Two techniques for sharing mobile-to-BS radio spectrum
- combined FDMA/TDMA: divide spectrum in frequency channels, divide each channel into time slots
- CDMA: code division multiple access



Summary

Approach	**SDMA**	**TDMA**	**FDMA**	**CDMA**
➢ Idea	▪ Segment ▪ space in to cells/ ▪ sectors	▪ Segment sending time into disjoint time-slots, ▪ demand driven or fixed patterns.	▪ Segment the frequency band into disjoint sub-bands	▪ Spread the spectrum using orthogonal codes
➢ Terminal	▪ Only one terminal can be active in one cell/one sector	▪ All terminals are active for short periods of time on the same frequency	▪ Every terminal ▪ has its own frequency, ▪ uninterrupted	▪ All terminals can be active at the same place at the same moment, ▪ uninterrupted
➢ Signal Separation	▪ Cell structure directed antennas	▪ Synchronization in the time domain	▪ Filtering in the frequency domain	▪ Code plus special receivers

Cont..

Advantage	▪ Very simple ▪ increases Capacity per km2	▪ Established, fully digital, ▪ very flexible	▪ Simple, established, ▪ robust	▪ Flexible, ▪ less planning needed, ▪ soft handover
Dis-advantage	▪ Inflexible, antennas typically fixed	▪ Guard space needed ▪ (multi-path propagation), ▪ Synchronization difficult	▪ Inflexible, ▪ frequencies are a scarce resource	▪ Complex receivers, ▪ needs more complicated power control for senders
Comment	▪ Only in combination with TDMA, FDMA or ▪ CDMA useful	▪ Standard in fixed networks, ▪ together with FDMA/SDMA used in many mobile networks	▪ Typically combined with ▪ TDMA (frequency hopping patterns) and ▪ SDMA (frequency reuse)	▪ Used in many 3G systems, ▪ higher complexity, ▪ lowered expectations; ▪ integrated with TDMA/FDMA

THANK YOU!!!

Network Architecture

- The main Components of a wireless System

I. Mobile Station (MS) : It incorporates

- user interface functions,
- radio functions, and
- control functions, with the most common equipment implemented in the form of a mobile telephone.

Cont..

II Base Station (BS)

a) **Base Transceiver Station (BTS):** The BTS consist of a radio equipment (transmitter and receiver - transceiver) and

- provides the radio coverage for a given cell or sector.

b) **Base Station Controller (BSC):**

- The BSC incorporates a control capability to manage one or more BTSs, executing the interfacing functions between BTSs and the network.

III. Mobile Switching Center (MSC)

- The MSC provides an automatic switching between users within the same network or other public switched networks, coordinating calls and routing procedures.

Telecommunication system

- Digital cellular networks are the segment of the market for mobile and wireless devices which are growing most rapidly.
- They are the wireless extensions of traditional Public Switched Telephone Network(PSTN)or Integrated service Digital Network (ISDN) networks
- Allow for seamless roaming with the same mobile phone nation or even worldwide.
- The first generation comprises analog systems, which typically rely on FDMA.
- The first 2G systems hit the market in the early nineties.

Cont..

- All these 2G systems introduced a TDMA mechanism in addition to FDMA, which is still used for channel separation.
- With cdma One the first CDMA technology was available in the US as a competitor to the TDMA technologies.

- Between the second and third generation there is no real revolutionary step.
- The systems evolved over time: GPRS introduced a packet-oriented service and higher data rates to GSM (but can also be used for TDMA systems in general).

Cont..

- Most, but not all, systems added CDMA technology to become 3G systems.
- GSM is the most successful digital mobile telecommunication system in the world today.
- The primary goal of GSM was to provide a mobile phone system that allows users to roam throughout Europe and provides voice services compatible to ISDN and other PSTN systems.
- GSM is a typical second generation system, replacing the first generation analog systems, but not offering the high worldwide data rates that the third generation systems, such as UMTS, are promising.

Performance characteristics of GSM

- **Communication**
  - mobile, wireless communication; voice and data services
- **Total mobility**
  - international access, chip-card enables use of access points of different providers
- **Worldwide connectivity**
  - one number, the network handles localization

Cont..

- **High capacity**
  - better frequency efficiency, smaller cells, more customers per cell
- **High transmission quality**
  - high audio quality and reliability for wireless, uninterrupted phone calls at higher speeds (e.g., from cars, trains)
- **Security functions** :
  - access control, authentication via chip-card and PIN

Disadvantages of GSM

- There is no perfect system!!
- no end-to-end encryption of user data
- no full ISDN bandwidth of 64 kbit/s to the user, no transparent B channel
- reduced concentration while driving
- electromagnetic radiation
- abuse of private data possible
- roaming profiles accessible
- high complexity of the system
- several incompatibilities within the GSM standards

GSM: Mobile Services

- GSM offers
  - several types of connections
    - voice connections, data connections, short message service
  - multi-service options (combination of basic services)
  - Three service domains
    1. Bearer Services
    2. Telematics Services
    3. Supplementary Services

I. Bearer Services

- Telecommunication services to transfer data between access points.
- Specification of services up to the terminal interface.
- Different data rates for voice and data (original standard).

II. Tele Services

- Telecommunication services that enable voice communication via mobile phones
- All these basic services have to obey cellular functions, security measurements etc.
- **Offered services**
  - mobile telephony
  - Emergency number
  - Multi-numbering
- **Additional services**
  - Non-Voice-Teleservices
  - Short Message Service (SMS)

III. Supplementary services

- Services in addition to the basic services, cannot be offered stand-alone
- Similar to ISDN services besides lower bandwidth due to the radio link
- May differ between different service providers, countries and protocol versions

Important services

- identification: forwarding of caller number
- suppression of number forwarding
- automatic call-back
- conferencing with up to 7 participants
- locking of the mobile terminal (incoming or outgoing calls)

Security in GSM

- **Security services**
- access control/authentication
  - ✓ user SIM (Subscriber Identity Module): secret PIN (personal identification number)
  - ✓ SIM   network: challenge response method

- **confidentiality**
  - voice and signaling encrypted on the wireless link (after successful authentication)

Cont..

- **anonymity**
  - temporary identity TMSI (Temporary Mobile Subscriber Identity)
    - newly assigned at each new location update (LUP)
  - encrypted transmission
- **3 algorithms specified in GSM**
  - A3 for authentication (“secret”, open interface)
  - A5 for encryption (standardized)
  - A8 for key generation (“secret”, open interface)

Data services in GSM

IS-136:

- It includes digital control channels (IS-54 uses analog AMPS control channels) and is more efficient.
- Fully digital phones can be used, several additional services are offered, e.g., voice mail, call waiting, identification, group calling, or SMS.
- IS-136 is also called North American TDMA (NA-TDMA) or Digital AMPS and operates at 800 and 1,900 MHz.
- Enhancements of D-AMPS/IS-136 toward IMT-2000 include advanced modulation techniques for the 30 kHz radio carrier, shifting data rates up to 64 kbit/s (first phase, called 136+).

Cont..

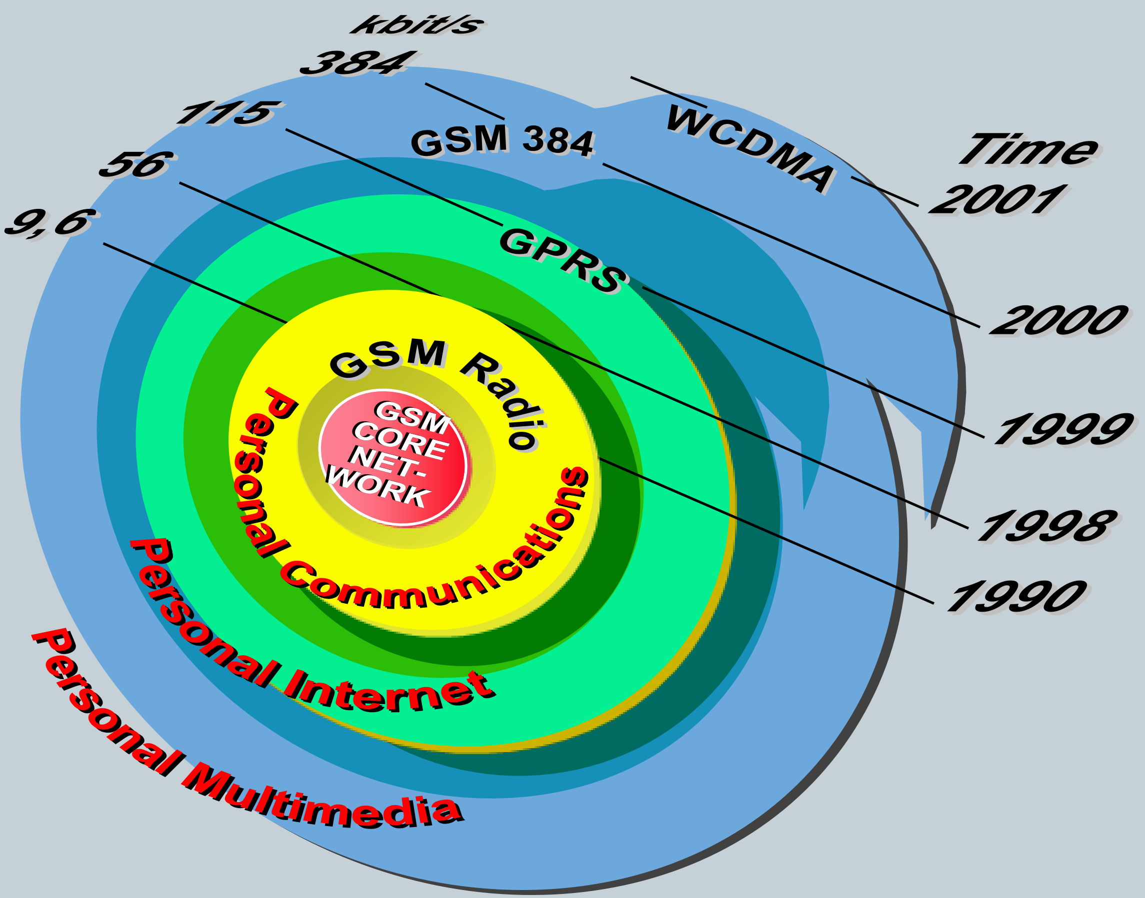
- The second phase, called 136HS (High Speed) comprises a new air interface specification based on the Enhanced data rates for global Evolution (EDGE) technology.
- IS-95 (promoted as cdma One) is based on CDMA, which is a completely different medium access method.
- Before deployment, the system was proclaimed as having many advantages over TDMA systems, such as its much higher capacity of users per cell, e.g., 20 times the capacity of AMPS..

Cont..

- IS-95 offers soft handover, avoiding the GSM Ping-Pong(senders want to receive delivery from the receiver) effect .
- However, IS-95 needs precise synchronization of all base stations (using GPS satellites which are military satellites, so are not under control of the network provider),
- frequent power control, and typically, dual mode mobile phones due to the limited coverage. The basic ideas of CDMA have been integrated into most 3G systems.

Data in GPRS Networks

GPRS



Cont..

- Part of GSM phase 2+
- General Packet Radio Service
  - General -> not restricted to GSM use (3rd generation systems)
  - Packet Radio -> enables packet mode communication over air
- Requires many new network elements into NSS
- Provides connections to external packet data networks (Internet, X.25)
- Main benefits
  - Resources are reserved only when needed and charged accordingly
  - Connection setup times are reduced
  - Enables new service opportunities

GPRS characteristics

GPRS uses packet switched resource allocation

- resources allocated only when data is to be sent/received

Flexible channel allocation

- one to eight time slots
- available resources shared by active users
- up and down link channels reserved separately
- GPRS and circuit switched GSM services can use same time slots alternatively

Traffic characteristics suitable for GPRS

- Intermittent, bursty data transmissions
- Frequent transmissions of small volumes of data
- Infrequent transmission of larger volumes of data

Cont..

- GPRS can interwork with GSM services through Gs-interface
- Most of the conventional GSM supplementary services are not applicable for GPRS
  - E.g., Call forwarding when busy, Calling line identification, Call waiting

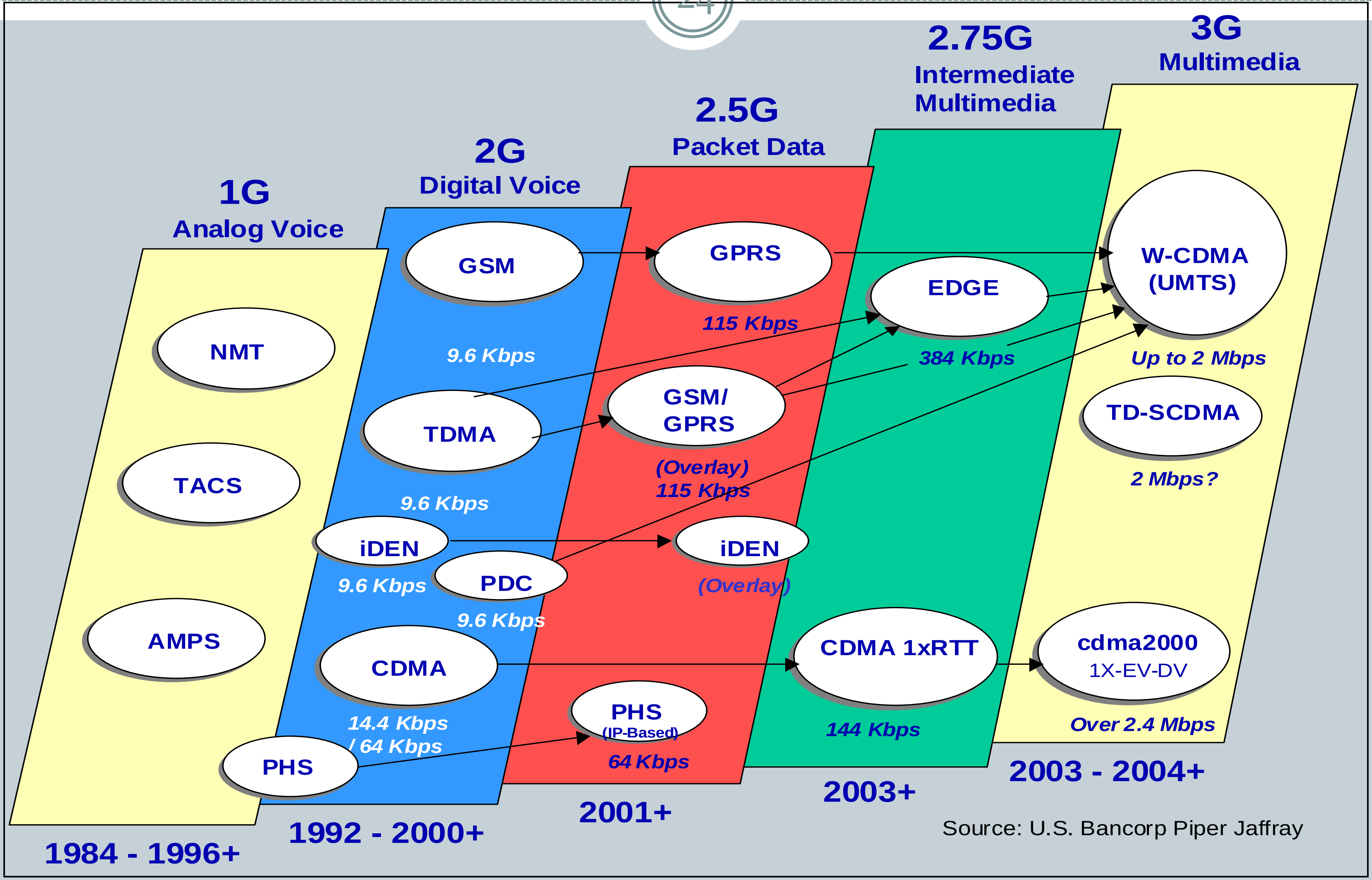
GPRS Standardization

- GPRS Phase 1:
  - Basic set of GPRS functionality
  - Optional features
- GPRS Phase 2: GPRS for Universal Mobile Telecommunication System(UMTS)
  - Certain issues are not included in the first release of the GPRS standard
  - New requirements have been pointed out for UMTS

UMTS

- Uses Wideband Code Division Multiple Access (W-CDMA)
- UMTS is a standard created for IMT-2000 (International Mobile Telecommunications-2000) by the ITU (International Telecommunication Union)
- Soft Handoffs
- Can connect with Packet or Circuit Switching
- Current coverage

Path to UMTS


1G
Analog Voice
NMT
TACS
AMPS
1984 - 1996+
2G
Digital Voice
GSM
9.6 Kbps
TDMA
9.6 Kbps
iDEN
9.6 Kbps
PDC
9.6 Kbps
CDMA
14.4 Kbps
/ 64 Kbps
PHS
1992 - 2000+
2.5G
Packet Data
GPRS
115 Kbps
GSM/
GPRS
(Overlay)
115 Kbps
iDEN
(Overlay)
PHS
(IP-Based)
64 Kbps
2001+
2.75G
Intermediate
Multimedia
EDGE
384 Kbps
CDMA 1xRTT
144 Kbps
2003+
3G
Multimedia
W-CDMA
(UMTS)
Up to 2 Mbps
TD-SCDMA
2 Mbps?
cdma2000
1X-EV-DV
Over 2.4 Mbps
2003 - 2004+
Source: U.S. Bancorp Piper Jaffray

UMTS Speed

- 384 kbps long range, 2 Mbps short range
- Theoretically can go up to 11 Mbps.

UMTS QoS Classes

- **Conversational class**:
  - Voice, video telephony, video gaming
- **Streaming class**:
  - Multimedia, video on demand, webcast
- **Interactive class**:
  - Web browsing, network gaming, database access
- **Background class**:
  - E-mail, short message service (SMS), file downloading

UMTS Problems

- Requires more battery power
- Can handoff UMTS to GSM, but not GSM to UMTS
- Initial poor coverage
- For video on demand and other bandwidth heavy services a Base Station needs to be set up every 1-1.5km. Unfeasible in rural areas.

END!!!

Wireless LAN

- A wireless LAN uses wireless transmission medium
- Used to have high prices, low data rates, occupational safety concerns, and licensing requirements
- Problems have been addressed
- Popularity of wireless LANs has grown rapidly.

Applications - LAN Extension

- Saves installation of LAN cabling
- Eases relocation and other modifications to network structure
- However, increasing reliance on twisted pair cabling for LANs
  - Most older buildings already wired with Cat 3 cable
  - Newer buildings are prewired with Cat 5

Cont..

- In some environments, role for the wireless LAN
  - Buildings with large open areas
    - ✓ Manufacturing plants, stock exchange trading floors, warehouses
    - ✓ Historical buildings
    - ✓ Small offices where wired LANs not economical
- May also have wired LAN
  - Servers and stationary workstations

Single Cell Wireless LAN Configuration

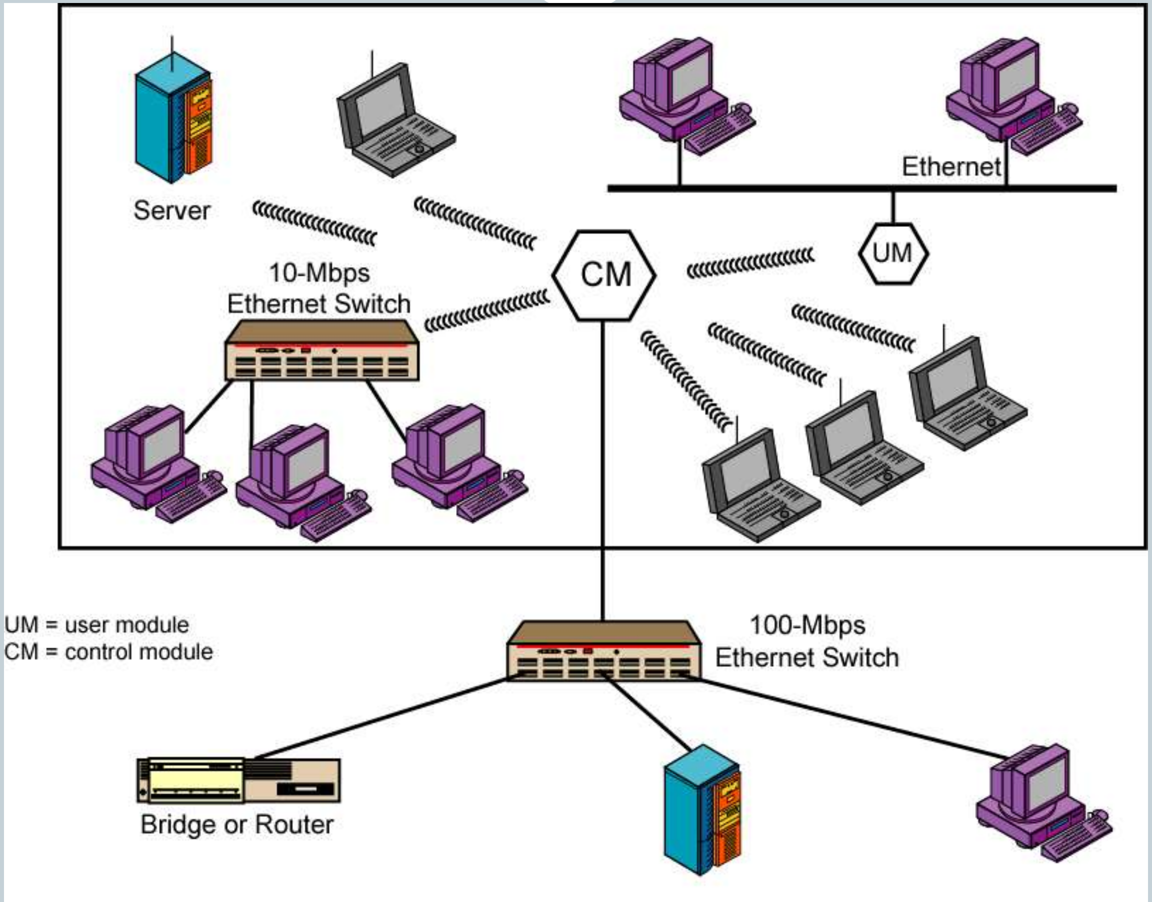


Figure 17.1 Example Single-Cell Wireless LAN Configuration

Multi-Cell Wireless LAN Configuration

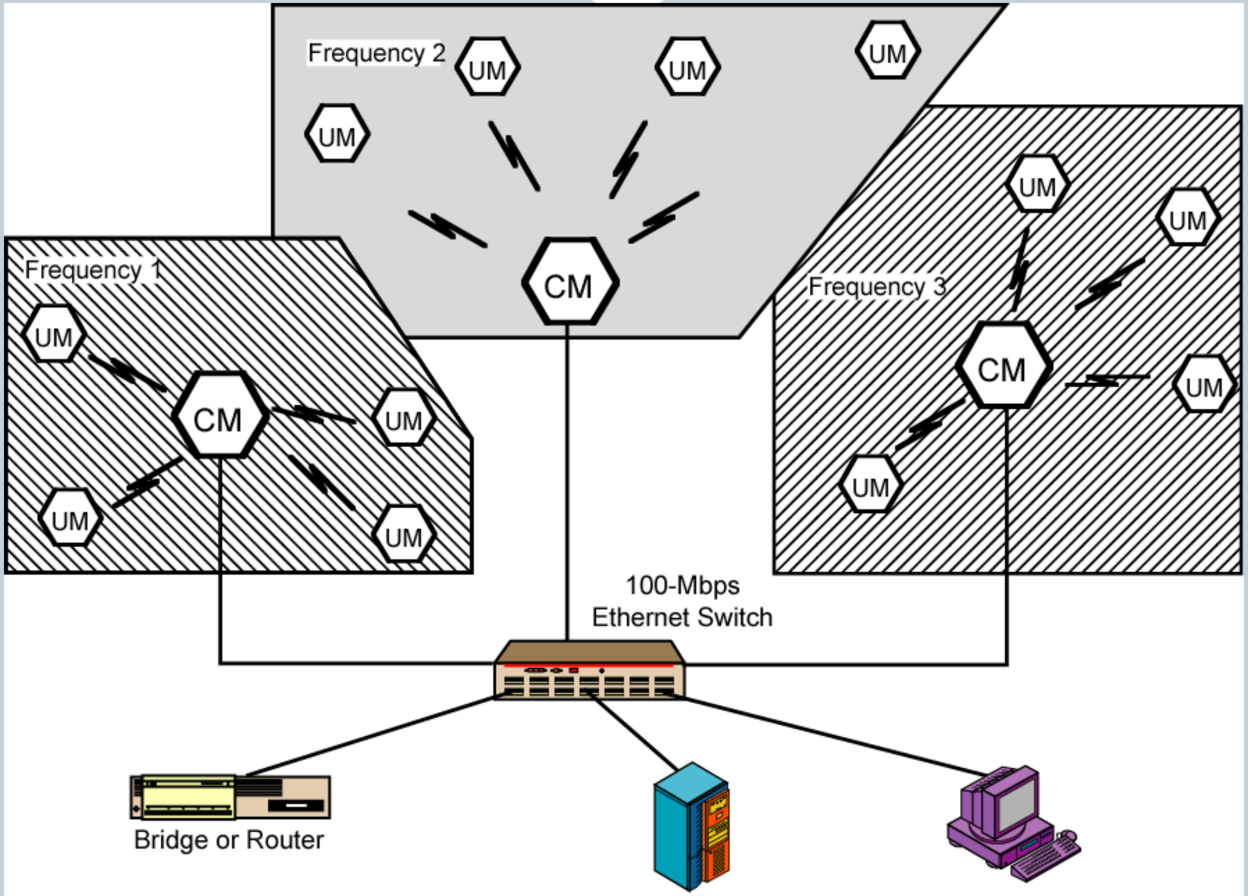


Figure 17.2 Example Multiple-Cell Wireless LAN Configuration

Applications-Cross-Building Interconnect

- Connect LANs in nearby buildings
- Point-to-point wireless link
- Connect bridges or routers

Applications - Nomadic Access

- Link between LAN hub and mobile data terminal
  - Laptop or notepad computer
  - Enable employee returning from trip to transfer data from portable computer to server
- Also useful in extended environment such as campus or cluster of buildings
  - Users move around with portable computers
  - May wish access to servers on wired LAN

Applications – Ad Hoc Networking

- Peer-to-peer network
- Set up temporarily to meet some immediate need
- E.g. group of employees, each with laptop or palmtop, in business or classroom meeting
- Network for duration of meeting

Add Hoc LAN

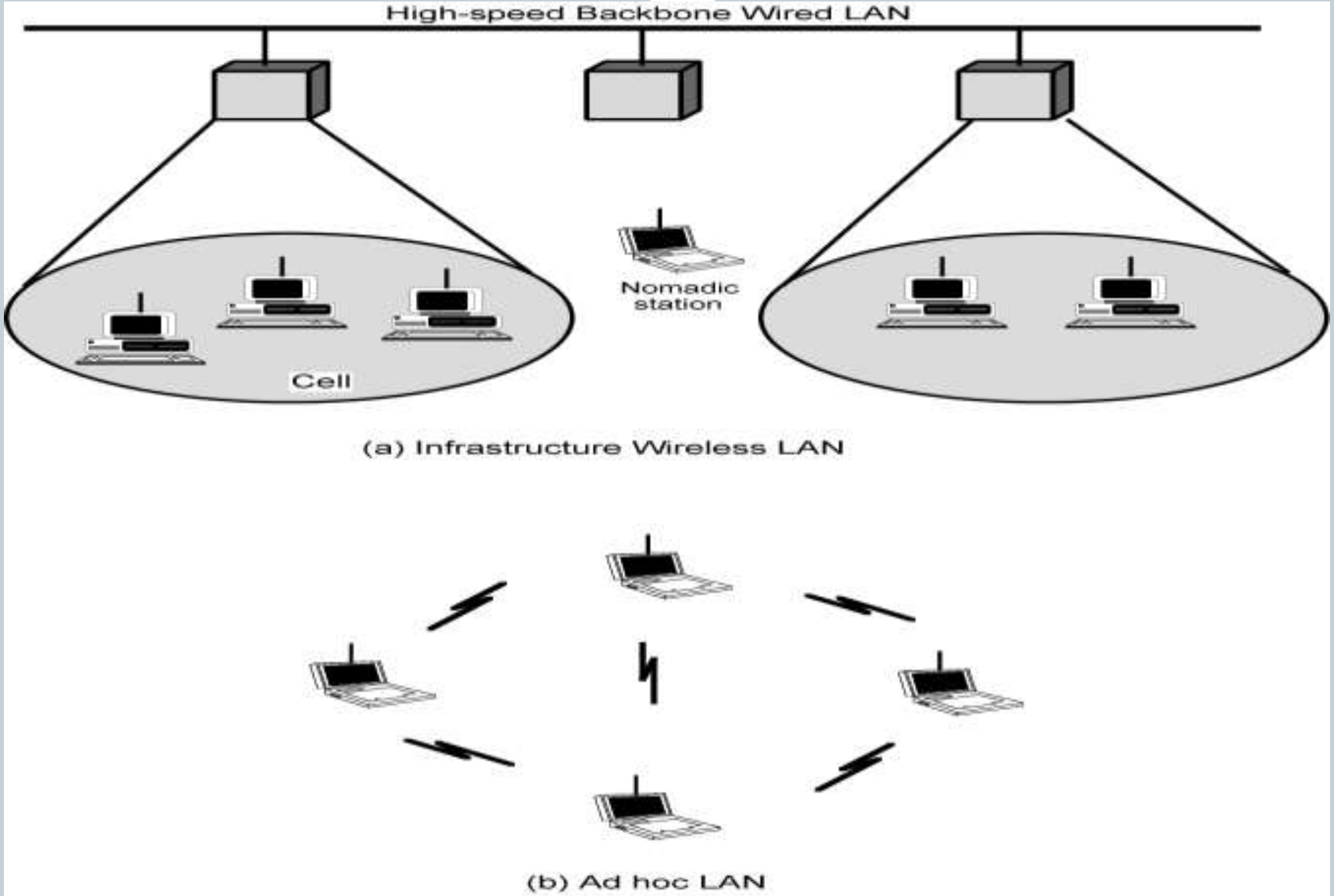


Figure 17.3 Wireless LAN Configurations

Wireless LAN Requirements

- Same as any LAN
  - High capacity, short distances, full connectivity, broadcast capability
- Throughput: efficient use wireless medium
- Number of nodes:Hundreds of nodes across multiple cells
- Connection to backbone LAN: Use control modules to connect to both types of LANs
- Service area: 100 to 300 m
- Low power consumption:Need long battery life on mobile stations
  - Mustn't require nodes to monitor access points or frequent handshakes

Cont..

- Transmission robustness and security: Interference prone and easily eavesdropped
- Collocated network operation: Two or more wireless LANs in same area
- Handoff/roaming: Move from one cell to another
- Dynamic configuration: Addition, deletion, and relocation of end systems without disruption to users.

IEEE 802.11 - BSS

- MAC protocol and physical medium specification for wireless LANs
- Smallest building block is basic service set (BSS)
  - Number of stations
  - Same MAC protocol
  - Competing for access to same shared wireless medium
- May be isolated or connect to backbone distribution system (DS) through access point (AP)
  - AP functions as bridge
- MAC protocol may be distributed or controlled by central coordination function in AP
- BSS generally corresponds to cell
- DS can be switch, wired network, or wireless network

BSS Configuration

- Simplest: each station belongs to single BSS
  - Within range only of other stations within BSS
- Can have two BSSs overlap
  - Station could participate in more than one BSS
- Association between station and BSS dynamic
  - Stations may turn off, come within range, and go out of range

Extended Service Set (ESS)

- Two or more BSS interconnected by DS
  - Typically, DS is wired backbone but can be any network
- Appears as single logical LAN to LLC

Wireless Technologies

- There are two technologies that have been developed as wireless cable replacements: Infrared (IRDA) and radio (Bluetooth).

- The name ‘Bluetooth’ was named after <u>10th century Viking king in Denmark Harald Bluetooth</u> who united and controlled Denmark and Norway.
- The name was adopted because Bluetooth wireless technology is expected to unify the telecommunications and computing industries.
- Who Started **<u>B</u>**luetooth?
- Bluetooth Special Interest Group (SIG)
- Founded in Spring 1998
- Now more than 2000 organizations joint the SIG

What Is Bluetooth?

- Bluetooth is an open standard for short-range digital radio to interconnect a variety of devices
  - Cell phones,
  - PDA,
  - notebook computers,
  - modems,
  - cordless phones,
  - pagers,
  - laptop computers,
  - printers,
  - cameras by developing a single-chip, low-cost, radio-based wireless network technology.

Bluetooth

- Simplifying communications between:

  - devices and the internet

  - data synchronization

- Operates in licensed exempt ISM band at 2.4ghz

- Uses frequency hoping spread spectrum

- Omni directional, no requiring line of sight

Bluetooth

- Bluetooth offers data speeds of up to 1 Mbps up to 10 meters (Short range wireless radio technology)
- Unlike IrDA, Bluetooth supports a LAN-like mode where multiple devices can interact with each other.
- The key limitations of Bluetooth are security and interference with wireless LANs.
- Short range wireless radio technology

Bluetooth

- ***Bluetooth is a Personal Area Network PAN Technology***
  - Offers fast and reliable transmission for both voice and data
  - Can support either one asynchronous data channel with up to three simultaneous synchronous speech channels
  - Support both packet-switching and circuit-switching

Bluetooth

- *Personal Area Network (PAN) Bluetooth is a standard that will*
  - Eliminate wires and cables between both stationary and mobile devices
  - Facilitate both data and voice communications
  - Offer the possibility of ad hoc networks and deliver synchronicity between personal devices

Bluetooth Topology

- Bluetooth-enabled devices can automatically locate each other
- Topology is established on a temporary and random basis
- Up to eight Bluetooth devices may be networked together in a master-slave relationship to form a Piconet (A collection of devices connected via Bluetooth technology in an ad hoc fashion).

Cont.

- One is master, which controls and setup the network
- All devices operate on the same channel and follow the same frequency hopping sequence
- Two or more piconet interconnected to form a scatter net
- Only one master for each piconet
- A device can’t be masters for two piconets
- The slave of one piconet can be the master of another piconet

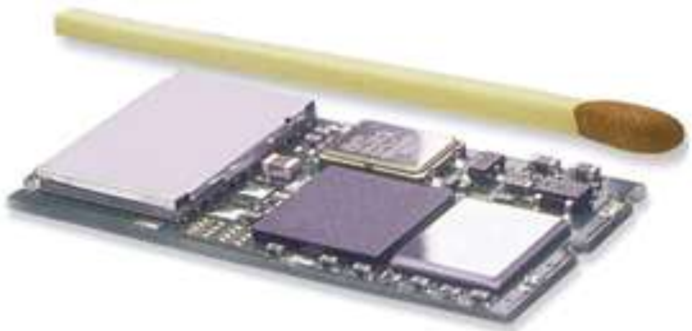
How Does It Work?

- Bluetooth is a standard for tiny, radio frequency chips that can be plugged into your devices

- These chips were designed to take all of the information that your wires normally send, and transmit it at a special frequency to something called a receiver Bluetooth chip.

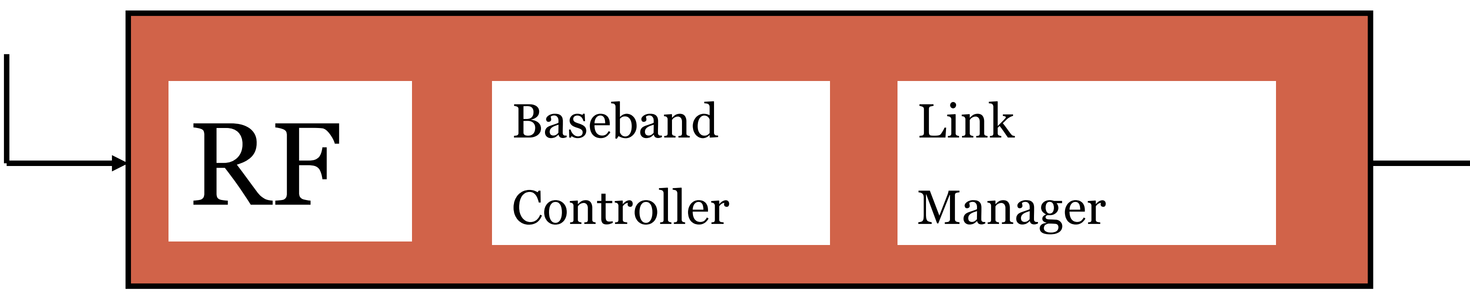
- The information is then transmitted to your device

Bluetooth Chip



Puce Bluetooth

Bluetooth Chip



Ad-hoc

- Is a network connection method which is most often associated with wireless devices.
- The connection is established for the duration of one session and requires no base station.
- Instead, devices discover others within range to form a network for those computers.
- Devices may search for target nodes that are out of range by flooding the network with broadcasts that are forwarded by each node.
- Connections are possible over multiple nodes (multi-hop ad hoc network).
- Routing protocols then provide stable connections even if nodes are moving around.

A piconet

- is an ad-hoc computer network of devices using Bluetooth technology protocols to allow one *master* device to interconnect with up to seven active *slave* devices
- Up to 255 further slave devices can be inactive, or *parked*, which the master device can bring into active status at any time.

A Typical Bluetooth Network Piconet

- **Master sends its globally unique 48-bit id and clock**
  - Hopping pattern is determined by the 48-bit device ID
  - Phase is determined by the master's clock
- **Why at most 7 slaves?**
  - (because a three-bit MAC address is used).
- **Parked and standby nodes**
  - Parked devices can not actively participate in the piconet but are known to the network and can be reactivated within some milliseconds
  - 8-bit for parked nodes
  - No id for standby nodes
  - Standby nodes do not participate in the piconet

Advantages (+)

- Wireless (No Cables)
- No Setup Needed
- Low Power Consumption (1 Milliwat)
- Industry Wide Support

Disadvantages (-)

- Short range (10 meters)
- Small throughput rates

  - Data Rate 1.0 Mbps
- Mostly for personal use (PANs)
- Fairly Expensive

Mobile Network Layer

Motivation for Mobile IP

- **Routing ‹**
  - Based on IP destination address, network prefix (e.g. 129.13.42) determines physical subnet ‹
  - To avoid an explosion of routing tables, only prefixes are stored and further optimizations are applied. ‹
  - Change of physical subnet implies change of IP address to have a topological correct address (standard IP) or needs special entries in the routing tables ‰

Cont..

- **Specific routes to end-systems? ‹**
  - change of all routing table entries to forward packets to the right destination
  - ‹does not scale with the number of mobile hosts and frequent changes in the location, security problems.

Motivation for Mobile IP

- ‰**Changing the IP-address?**
- ‹adjust the host IP address depending on the current location
  - „Assigning a new IP address
- Problem: nobody knows about this new address. ‹
- Almost impossible to find a mobile system, Domain Name System (DNS) updates take to long time. ‹
- Transmission control protocol (TCP) connections break, security problems „
  - TCP connection = {source IP, source port, destination IP, destination port} „
  - TCP connection cannot survive any address change.

Requirements to Mobile IP

- **%oTransparency ‹**
  - Mobility should remain 'invisible' for many higher layer protocols and applications ‹For TCP, mobile computer must keep its IP address. ‰
- **Compatibility ‹**
  - support of the same layer 2 protocols as IP ‹no changes to current end-systems and routers required ‹mobile end-systems can communicate with fixed systems ‰
- **Security ‹**
  - The minimum requirement: all the messages related to the management of Mobile IP are authenticated. ‰
- **Efficiency and scalability ‹**
  - Only little additional messages to the mobile system required (connection typically via a low bandwidth radio link)

Mobile IP Design Goals

- A mobile node must be able to communicate with other nodes after changing it’s link-layer attachment, yet without changing its IP address ‰
- A mobile node must be able to communicate with other nodes that do not implement mobile IP ‰
- Mobile IP must use authentication to offer security against redirectment attacks ‰
- The number of administrative messages should be small to save bandwidth & power ‰
- Mobile IP must impose no additional constraints on the assignment of IP addresses

Mobile IP and IPv6

- %Mobile IP was developed for IPv4, but IPv6 simplifies the protocols ‹
  - Security is integrated and not an add-on, authentication of registration can be assigned via auto-configuration.
  - Every node has address auto-configuration ‹
  - No need for a separate FA, all routers perform router advertisement which can be used instead of the special agent advertisement.
  - addresses are always co-located ‹

IP version 6 (Mobile IP)

Mobile IPv4	Mobile IPv6
Mobile node, home agent, home link, foreign link	(same)
Mobile node’s home address	Globally routable home address and link local home address.
Foreign agent	A “plain” IPv6 router on the foreign link(foreign agent no longer exists)
Foreign agent care-of address	All care-of addresses are collocated
Collocated care-of address	
Care-of address obtained via Agent Discovery, DHCP, or manually	Care-of address obtained via Stateless Address auto-configuration or manually
Agent Discovery	Router Discovery
Registration with home agent	Notification of home agent and other correspondents

Mobility Classification

- ‰**Micro-mobility** is the movement of an MN within or across different BSs within a subnet and occurs very rapidly. (local mobility) ‰
- **Macro-mobility** is the movement of an MN across different subnet within a single domain or region, and occurs relatively less frequently. (intra-domain mobility).‰
- **Global Mobility** is the movement of an MN among different administrative domains or geographical regions. (inter-domain mobility).

Mobile Ad-hoc network

What is a MANET?

- MANET – (Mobile Ad-Hoc NET work) a system of mobile nodes (laptops, sensors, etc.) interfacing without the assistance of centralized infrastructure (access points, bridges, etc.)

Ad-hoc network

Why Ad Hoc Networks ?

- Setting up of fixed access points and backbone infrastructure is not always viable
  - Infrastructure may not be present in a disaster area or war zone
  - Infrastructure may not be practical for short-range radios; Bluetooth (range ~ 10m)
- Ad hoc networks
  - Do not need backbone infrastructure support
  - Are easy to deploy
  - Useful when infrastructure is absent, destroyed or impractical

Ad-hoc Network problem

- Unstable paths
- Processing power
- Battery life
- Time delays
- High cost of memory

Factors Affecting MANETs

- ❖ Scalability
- ❖ Power vs. Latency
- ❖ User Education
- ❖ Security
- ❖ Incompatible Standards
- ❖ Data Rates
- ❖ Coverage

Many Applications

- Personal area networking
  - cell phone, laptop, ear phone, wrist watch
- Military environments
  - soldiers, tanks, planes
- Civilian environments
  - taxi cab network
  - meeting rooms
  - sports stadiums
  - boats, small aircraft
- Emergency operations
  - search-and-rescue
  - policing and fire fighting

END!!!